



STRICTLY CONFIDENTIAL

PREPARED FOR

PDF Audit 1780600309928

50 kW

COMMERCIAL SOLAR PLANT

PDF Audit 1780600309928

Bhopal, MP On-Grid Commercial Solar 50 KW · 93 PANELS · 100% LOAD COVERAGE

KEY PERFORMANCE INDICATORS

ANNUAL SAVING

₹5.1L

NET INVESTMENT

₹42.5L

PAYBACK

8.3 yr

25-YEAR ROI

12.0%

25-YEAR PROFIT

₹2.4Cr

Green Campus & Sustainability Impact

Solar is not only a cost-saving investment — it is a branding, CSR, and future-ready campus asset



GREEN CAMPUS RECOGNITION

Position your institution as a future-ready Green Campus and demonstrate environmental leadership to students, parents, and the community.

SUSTAINABILITY BADGE - GOLD



ANNUAL CO₂ REDUCTION

47.1 t

vs grid electricity



TREE EQUIVALENT (25 YR)

53,568

CO₂ absorption impact



CLEAN ENERGY GENERATED

57.5

MWh/yr · on-site solar output



ENVIRONMENTAL IMPACT SCORE

100/100

campus sustainability index

25-YEAR ENVIRONMENTAL IMPACT

1179 t

CO₂ emissions avoided

1437.2 MWh

clean energy produced

53,568

tree-years equivalent

Why Schools Achieve Excellent Solar Utilization



School operating hours (8 AM – 4 PM) overlap with peak solar generation hours

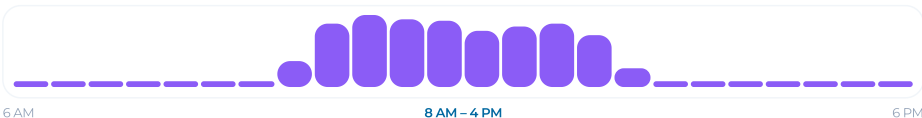


Maximum daytime consumption — directly offset by on-site solar

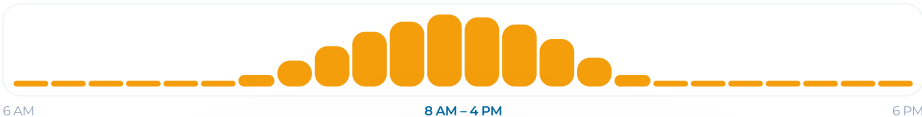


Reduced grid dependence · faster ROI · predictable energy costs

8 AM → 4 PM SCHOOL USAGE



SOLAR PRODUCTION CURVE



50 kW proposed system · ~100% daytime load coverage

Solar as a Learning Asset

The solar plant becomes a real-world learning laboratory for energy, sustainability, and performance

The solar plant becomes a real-world learning laboratory where students can observe energy generation, sustainability metrics, and renewable energy performance.



Real-Time Generation Monitoring

View daily, weekly, and monthly production on a live dashboard — ready for classroom discussions on energy data.



Science & STEM Learning

Photovoltaic effect, DC/AC conversion, and energy conservation — practical examples aligned with school science curricula.



Renewable Energy Awareness

Real campus data for eco-clubs, assembly talks, and sustainability weeks — empowering student-led green initiatives.



Student Sustainability Projects

Carbon footprint calculators, science fair models, and inter-house green competitions — live solar metrics available for all.



MONITORING PLATFORM

Estimated 57.5 MWh/yr from the proposed 50 kW system — live data available to school staff and students via IoT data logger and OEM cloud portal.

DCR COMPARISON

DCR vs Non-DCR — Cost Impact

Waaree 540W modules — subsidy & ALMM impact included

DCR / ALMM Higher subsidy potential ₹21,09,240 ₹42/Wp · 93 modules	NON-DCR Lower module rate ₹19,08,360 ₹38/Wp · 93 modules
 DCR premium ₹2,00,880 (+11%) panel hardware vs Non-DCR	DCR / ALMM-listed modules may qualify for higher central subsidy under applicable schemes.

Plant Capacity Scenarios

Alternative kW options alongside our recommended sizing

Recommended

☆ Recommended

50 kW

Net investment	₹42,50,000
Annual generation	57,488 kWh
Annual saving	₹5,10,000
Payback	8.3 yr
Roof area (est.)	~325 m²

Conservative

40 kW

Net investment	₹34,00,000
Annual generation	45,990 kWh
Annual saving	₹4,08,000
Payback	8.3 yr
Roof area (est.)	~260 m²

Expansion

60 kW

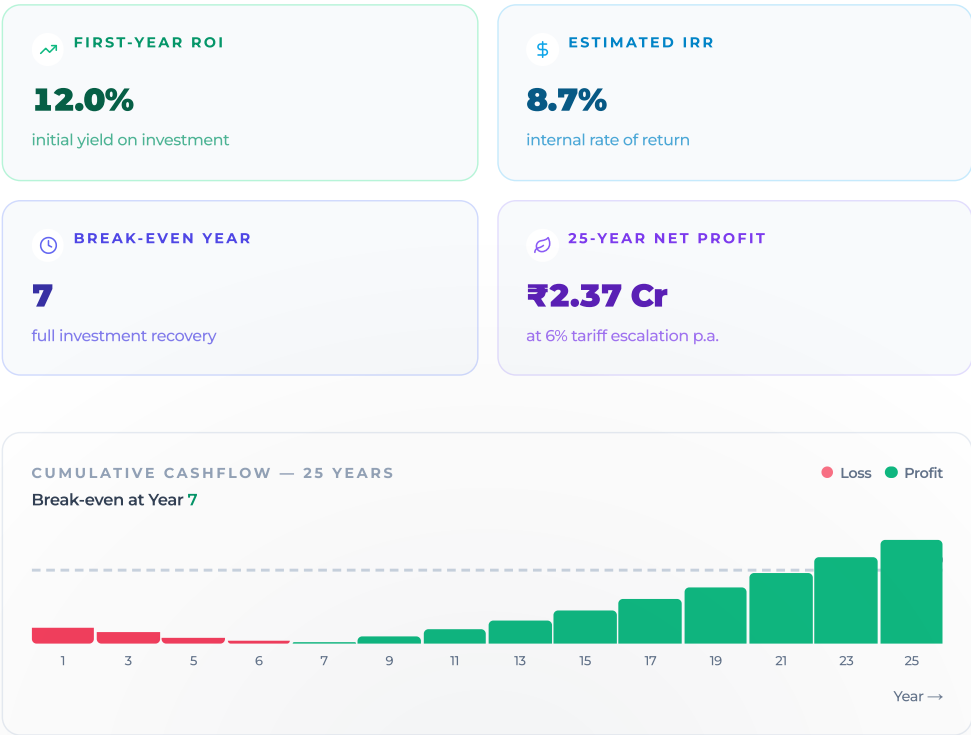
Net investment	₹51,00,000
Annual generation	68,986 kWh
Annual saving	₹6,12,000
Payback	8.3 yr
Roof area (est.)	~390 m²

0 2

RETURN ON INVESTMENT

Investment Analysis Dashboard

Full financial return profile — payback, IRR, and 25-year wealth creation



25-YEAR ENERGY COST COMPARISON



NET LIFECYCLE SAVING

₹2.39 Cr

grid vs solar over 25 years

ENVIRONMENTAL ROI

47.1_T

CO₂ Avoided/yr

53568

Tree Equivalent

1179_T

25yr CO₂

25-Year Economic Analysis

Rising grid tariffs (6% p.a.) vs flat solar cost — detailed year-by-year cashflow

FIRST-YEAR ROI

12.0%

annualised

ESTIMATED IRR

8.7%

Rule-of-72 basis

LCOE

₹2.96/kWh

levelised cost of energy

NPV (@ 10%)

₹34.5 L

positive

YEAR-BY-YEAR CASHFLOW STATEMENT			
YEAR	GRID COST	SOLAR SAVING	CUMULATIVE
Yr 1	₹6.0 L	₹5.1 L	₹-37.4 L
Yr 2	₹6.4 L	₹5.4 L	₹-32.0 L
Yr 3	₹6.7 L	₹5.7 L	₹-26.3 L
Yr 4	₹7.1 L	₹6.1 L	₹-20.2 L
Yr 5	₹7.6 L	₹6.4 L	₹-13.8 L
Yr 10	₹10.1 L	₹8.6 L	+₹24.7 L
Yr 15	₹13.6 L	₹11.5 L	+₹76.2 L
Yr 20	₹18.2 L	₹15.4 L	+₹1.45 Cr
Yr 25	₹24.3 L	₹20.6 L	+₹2.37 Cr
* Assumes 6% annual electricity tariff escalation			

CUMULATIVE PROFIT TRAJECTORY



BREAK-EVEN SUMMARY

Full investment recovered in Year 7

Then **₹2.37 Cr** net profit over 25 years

Technical Design Basis

Key engineering parameters governing system sizing, layout, and performance estimates

SYSTEM DESIGN PARAMETERS

DC Capacity (STC)	50.22 kWp	AC Capacity (inverter rated)	50 kW
DC/AC Ratio	1	Peak Sun Hours (India avg)	5.0 hrs/day
Performance Ratio (PR)	78%	Capacity Factor (CF)	13.1%
Specific Yield	1150 kWh/kWp/yr	Annual Generation	57.5 MWh/yr
Panel Efficiency (STC)	~25.7%	Module Wattage	540 Wp per panel
Panel Count	93 nos	Load Coverage	100%

STANDARDS & COMPLIANCE

- IEC 61215 / BIS certified PV modules
- MNRE-ALMM listed panel brand
- IEC 62109 certified string inverter
- IS 13947 / IEC 60947 switchgear
- IS/IEC 62446 commissioning test
- IS 732 electrical installation wiring
- CEA Safety Regulations 2010
- Net-metering per SERC regulations

INSTALLATION & COMMISSIONING PROCESS

01

Site Survey

Shading analysis · Load profiling · Structural assessment

02

System Design

PVsyst simulation · SLD drawing · Structural design

03

Procurement

ALMM-listed panels · Certified inverter · BOS materials

04

Installation

Certified engineers · Safe wiring · Grid interconnect

05

Testing & Commissioning

IV curve · Insulation test · Commissioning report

06

Net-Metering

DISCOM application · Meter installation · Grid export

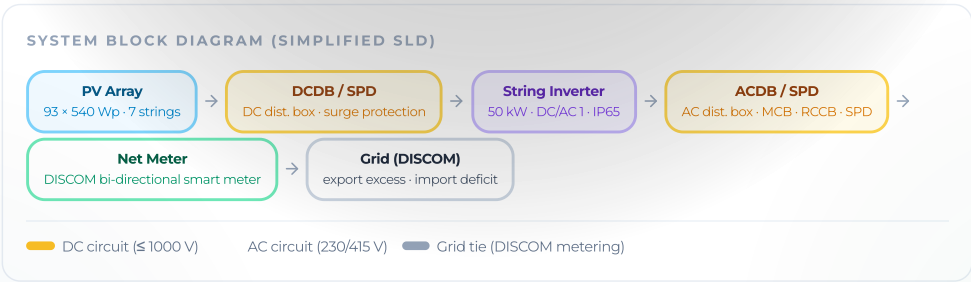
SCHOOL SAFETY

Designed for Safe Educational Environments

- ⚡ Child-safe engineering practices
- ⚡ Surge protection (SPD Type 2)
- ⚡ Remote monitoring & alerts
- ⚡ Certified BIS / IEC components
- ⚡ Proper earthing per IS 3043
- ⚡ Low-maintenance · safe operation
- 👁 All AC/DC panels lockable · roof access restricted · post-commissioning safety walkthrough included

System Component Architecture

DC string to grid — single-line diagram overview with component specifications



COMPONENT SPECIFICATIONS				
COMPONENT	SPECIFICATION	BRAND / STANDARD	QTY	WARRANTY
Solar PV Modules	540 Wp Mono PERC Half-cut	JSW	93 nos	25 yr linear power
String Inverter	50 kW Grid-Tie, IP65	Growatt / Deye (BIS-certified)	4 nos	5 yr standard
Mounting Structure	Hot-dip galvanised MS, wind-load engineered	IS 2062 steel	50 kW capacity	10 yr structural
DCDB / ACDB	IP54 weatherproof, SPD + MCB + Isolator	Havells / Schneider / Legrand	1 set each	1 yr product
DC Cables	4 mm ² / 6 mm ² Solar DC (TÜV-certified)	Havells / Polycab	Per layout design	25 yr UV rating
Earthing System	IS 3043 — 4-electrode GI pipe earthing	Copper-bonded / GI	As per design	Lifetime

PROTECTION PROVISIONS

- DC circuit breakers + fuses (per string)
- Type-2 surge protection — DC & AC side
- RCCB + RCBO for AC installation
- Insulation monitoring relay (IMR)
- IS 3043 compliant 4-electrode earthing
- Anti-islanding protection per CEA

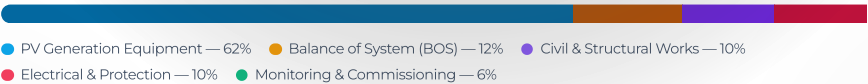
MONITORING & CONTROL

- Inverter cloud portal via Wi-Fi / LAN
- Real-time generation + consumption tracking
- String-level performance data
- Fault alarm + SMS / email alerts
- Mobile app — iOS & Android
- Monthly performance report (PDF)

Tiered Bill of Materials

Complete EPC supply-and-install scope — all categories with specifications, makes, and warranties

APPROXIMATE COST DISTRIBUTION



Total project value: ₹42.5 L · Net after subsidy: ₹42.5 L

DCR vs Non-DCR Panel Pricing 93 modules · 540W

DCR PANEL
ALMM listed · Subsidy eligible
₹42.0/Wp
₹21.09 L total
PM Surya Ghar subsidy applicable

NON-DCR PANEL
Import / non-ALMM
₹38.0/Wp
₹19.08 L total
No subsidy · Lower upfront cost

⇄ **DCR Premium** **+₹4.0/Wp**
+₹201k total · offset by subsidy benefit

PV GENERATION EQUIPMENT

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Solar PV Modules	540 Wp Mono PERC Half-cut, $\eta \geq 20.8\%$	JSW	93	nos	25 yr linear
String Inverter (Three-phase)	50 kW on-grid, Three-phase, MPPT, IP65, LCD	Growatt / Deye (BIS-certified)	1	nos	5 yr standard

BALANCE OF SYSTEM (BOS)

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Solar DC Cable (4 mm ²)	TÜV 2Pfg 1169, UV-rated, 1 kV	Polycab / Havells	400	metres	25 yr UV
AC Power Cable (16 mm ²)	XLPE armoured, IS 1554	Finolex / KEI	200	metres	10 yr
MC4 Connectors	IP68, TÜV certified, UV-stable	Stäubli / Amphenol	28	pairs	25 yr
Cable Management (HDPE conduit)	25 mm / 40 mm IS 9537, UV-rated	Finolex / Sudhakar	As per layout	lot	10 yr

CIVIL & STRUCTURAL WORKS

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Module Mounting Structure	Hot-dip galvanised MS, wind-load designed	IS 2062, zinc $\geq 85 \mu\text{m}$	50 kW	capacity	10 yr structural
Foundation / Anchor bolts	M12/M16 SS 304, chemical anchor	Hilti / Fischer equiv.	186 approx	nos	10 yr
Cable Tray / Ladder	GI 2.0 mm perforated, IS 2062	ABB / Legrand equiv.	As per layout	lot	10 yr

● ELECTRICAL & PROTECTION

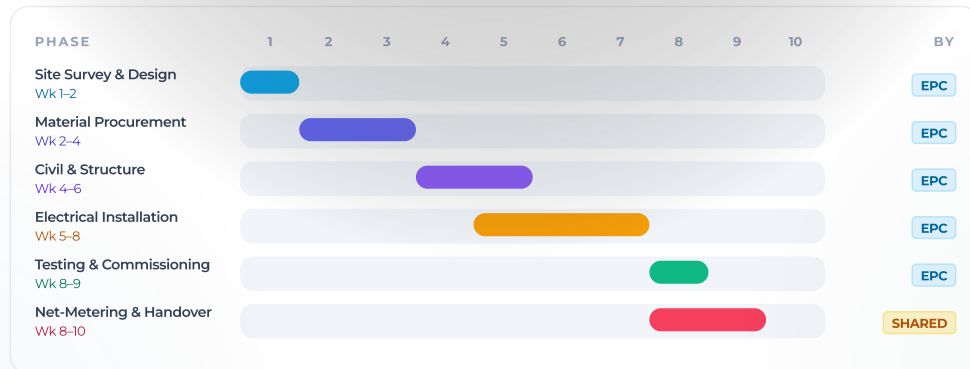
ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
DCDB (DC Distribution Box)	IP54, DC MCB + fuse + SPD Type 2	Havells / Schneider	1	nos	1 yr
ACDB (AC Distribution Box)	IP54, MCB + RCCB + SPD + energy meter	Havells / Schneider	1	nos	1 yr
Earthing System	IS 3043 — 4-electrode GI pipe, 3 m deep	Copper-bonded / GI	4 nos	electrodes	Lifetime
Lightning Arrestor	IEC 62305 — Class I/II, roof-mount	OBO Bettermann / equiv.	1 set	lot	5 yr

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Smart Monitoring Device	IoT data logger, Wi-Fi + Ethernet	Inverter OEM cloud portal	1	lot	2 yr
Net Energy Meter (NEM)	Bi-directional, CT/PT type per DISCOM spec	DISCOM approved make	1	nos	DISCOM owned
Commissioning & Testing	IV curve · insulation test · IS/IEC 62446 report	In-house certified engineers	1	lot	N/A
DISCOM Net-Metering Approval	Application + inspection + meter install support	EPC scope	1	application	N/A

⌵ This BOM reflects indicative specifications. Final BOM will be provided post site survey and detailed design. Equivalent brand substitutions may be used with prior written approval.

Project Execution Timeline

Estimated 10-week project plan for 50 kW on-grid system



01 Site Survey & Design

- Shadow & tilt analysis
- Load profiling & tariff study
- PVsyst / SketchUp modelling
- SLD + structural drawing

Design approved

02 Material Procurement

- PO to panel & inverter vendors
- BOS items sourcing
- Logistics to site

Materials at site

03 Civil & Structure

- Structural foundation
- Module mounting structure erection
- Cable trays & conduit

Structure complete

04 Electrical Installation

- Panel mounting & wiring
- DCDB / ACDB installation
- Inverter & earthing
- AC interconnection

Pre-commissioning ready

05 Testing & Commissioning

- IV curve test
- Insulation resistance test
- System energisation
- IS/IEC 62446 commissioning report

System live

06 Net-Metering & Handover

- DISCOM NEM application
- Meter inspection & approval
- Training — customer team
- Document handover

Project closed

Smart Monitoring & Annual Maintenance

24×7 real-time data, proactive fault management, and long-term performance assurance

SMART MONITORING CAPABILITIES

-  **Real-Time Generation**
Minute-level kW output & kWh energy tracking
-  **String-Level Monitoring**
Per-string performance & mismatch detection
-  **Grid Export / Import**
Net-meter data + consumption analysis
-  **Fault Alarms & Alerts**
SMS / email — fault, low production, grid error
-  **Mobile App**
iOS & Android — live dashboard & history
-  **Monthly Performance Report**
PDF report — generation, saving, CO₂ data
-  **Historical Data**
5+ year historical data storage on cloud
-  **Remote Diagnostics**
Remote firmware update & inverter diagnostics

SERVICE LEVEL COMMITMENTS (SLA)

METRIC	STANDARD	PREMIUM
Response Time (remote)	4 business hrs	2 business hrs
On-site Visit (fault)	48 hrs	24 hrs
Inverter Downtime Cap	72 hrs max	48 hrs max
System Uptime Guarantee	≥ 95%	≥ 98%
Annual Service Visits	1 per year	2 per year
Report Delivery	Monthly	Weekly + Monthly

ANNUAL MAINTENANCE CONTRACT — PLAN COMPARISON

1 Year

Complimentary

First year — complimentary

- ✔ Annual inspection visit
- ✔ Panel cleaning (1 visit)
- ✔ Inverter firmware update
- ✔ Terminal tightening check
- ✔ Performance report
- ✖ Quarterly panel cleaning
- ✖ Spare parts coverage

RECOMMENDED

5 Year

₹85 k

15% discount — 5-year package

- ✔ Annual inspection visit
- ✔ Panel cleaning (2× / yr)
- ✔ Inverter firmware update
- ✔ Terminal tightening check
- ✔ Performance report
- ✔ Quarterly panel cleaning
- ✖ Spare parts coverage

10 Year

₹1.5 L

25% discount — 10-year package

- ✔ Annual inspection visit
- ✔ Panel cleaning (2× / yr)
- ✔ Inverter firmware update
- ✔ Terminal tightening check
- ✔ Performance report
- ✔ Quarterly panel cleaning
- ✔ Spare parts coverage

Commercial & Contractual Terms

Payment schedule, warranty matrix, EPC scope, and key commercial terms

 **PAYMENT SCHEDULE**

Total Project Value

₹42.5 L

1

advance

₹10.6 L (25%)

2

material

₹21.3 L (50%)

3

installation

₹8.5 L (20%)

4

commissioning

₹2.1 L (5%)

 **KEY COMMERCIAL TERMS**

Quotation Validity	30 days from date of issue
Payment Terms	Milestone-based (see above)
Project Completion	10 weeks from WO
Price Basis	Inclusive of current GST + freight
GST Rate	12% (solar energy equipment)
Dispute Resolution	Arbitration Act 1996, India

 **EPC SCOPE SUMMARY**

INCLUDES

- All EPC supply and installation works
- DISCOM NEM application & liaison support
- IS/IEC 62446 commissioning test report
- Cloud monitoring portal activation
- Operations & maintenance training
- First-year AMC (complimentary)

EXCLUDES

- DISCOM net meter cost (DISCOM-owned)
- Internal load re-cabling / DB upgrade
- Roof waterproofing / civil repairs
- Electricity connection augmentation
- PM Surya Ghar subsidy processing

 WARRANTY MATRIX

COMPONENT	DURATION	WARRANTOR	COVERAGE
Solar PV Modules — Product	10 years	Manufacturer	Manufacturing defects
Solar PV Modules — Power Output	25 years	Manufacturer	Linear degradation $\leq 0.45\%/yr$
String Inverter	5 years	Manufacturer	Standard product warranty
Mounting Structure	10 years	EPC / Fabricator	Corrosion & structural integrity
Electrical Workmanship	2 years	EPC	Installation workmanship guarantee
Monitoring System	2 years	OEM / EPC	Hardware & software
* Contact Harihar Solar for warranty claims. All warranties subject to normal maintenance and use conditions.			

Accept This Proposal

Decision summary, acceptance section, and next steps to commission your solar system

EXECUTIVE DECISION SUMMARY		
50 kW On-Grid Solar System — PDF Audit 1780600309928		
NET INVESTMENT	ANNUAL SAVING	BREAK-EVEN
₹42.5 L	₹5.1 L	7 yr
post-subsidy	from year one	full recovery
25-YEAR ROI	IRR (EST.)	25-YEAR PROFIT
12%	8.7%	₹2.37 Cr
annualised	internal rate	net profit

COST OF DELAY

Every month without solar costs ₹43 k in foregone savings.
A 3-month delay = ₹1.3 L lost permanently.

PROPOSAL ACCEPTANCE

By accepting this proposal, you agree to the EPC scope and commercial terms described above. Work order to be issued to proceed.

Client Signature
Name & Designation

EPC Authorised Signatory
Harihar Solar

Date

Place

Proposal Ref: 1BC0491C

CONTACT YOUR EPC PARTNER



Harihar Solar

100% Local · Satna · Madhya Pradesh

☎ +91-9993322267 · harihar@solar.com

✉ Contact for email

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This 50 kW system will offset 1179 tonnes of CO₂ over 25 years.
Equivalent to planting 53568 trees.